

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 42

WORLD CLIMATIC TYPES
THE SIBERIAN CLIMATE

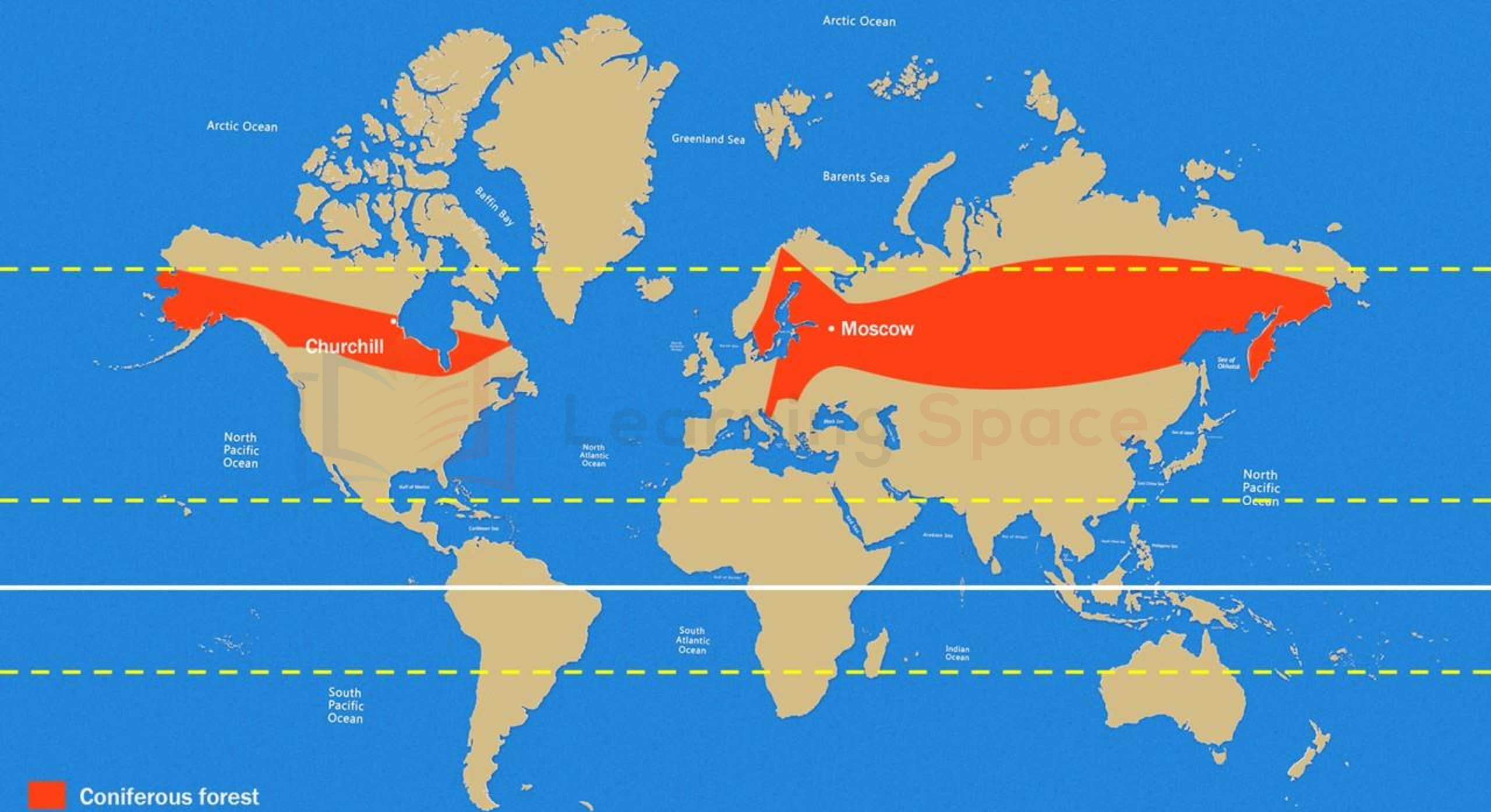
WORLD CLIMATIC TYPES

7

THE COOL TEMPERATE CONTINENTAL (SIBERIAN) CLIMATE

DISTRIBUTION

- 1 The Siberian Climate is experienced only in the northern hemisphere where the continents within the **high latitudes** have a **broad east-west spread**.
- 2 On its poleward side, it merges into the **Arctic tundra of Canada and Eurasia** at **around the Arctic Circle**, Southwards, the climate becomes less severe and **fades** into the **temperate Steppe climate**.
- 3 The predominant vegetation of this **Siberian or "Sub-Arctic" type of climate** is **evergreen coniferous forest**.
- 4 It stretches in a great, continuous belt across North America, Europe and Asia.
- 5 The greatest single band of the **coniferous forest** is the **taiga**.
- 6 The Siberian Climate is conspicuously absent in the southern hemisphere **because of the narrowness of the southern continents in the high latitudes**.
- 7 Also, the **strong oceanic influences of the southern hemisphere** further reduces the severity of the winter and the extent of this climate.



7

THE SIBERIAN CLIMATE

CLIMATE

a

TEMPERATURE



1

The climate of the Siberian type is characterized by a **bitterly cold winter of long duration**, and a **cool brief summer**.

2

Spring and autumn are merely brief transitional periods.

3

The **extremes of temperature** are so great in Siberia that it is often referred to as the “**Cold pole of the Earth**”.

7 THE SIBERIAN CLIMATE

CLIMATE

a TEMPERATURE



4 Some of the lowest temperatures in the world are recorded in Verkhoyansk (68°N 113°E and only 330 feet in altitude) where -68°C was once recorded.

5 With such low temperatures in the cold season, heavy snowfall can be expected.

6 Frosts occur as early as August and by September lakes and ponds are already ice-bound. All over Russia, nearly all the rivers are frozen for this part of the year.

7 THE SIBERIAN CLIMATE

CLIMATE

a TEMPERATURE



7 In normal years, the Volga is ice-covered for about 150 days, while those further north (e.g. the lower courses of the Ob, Lena and Yenisey) are ice-covered for more than 210 days or 7 months!

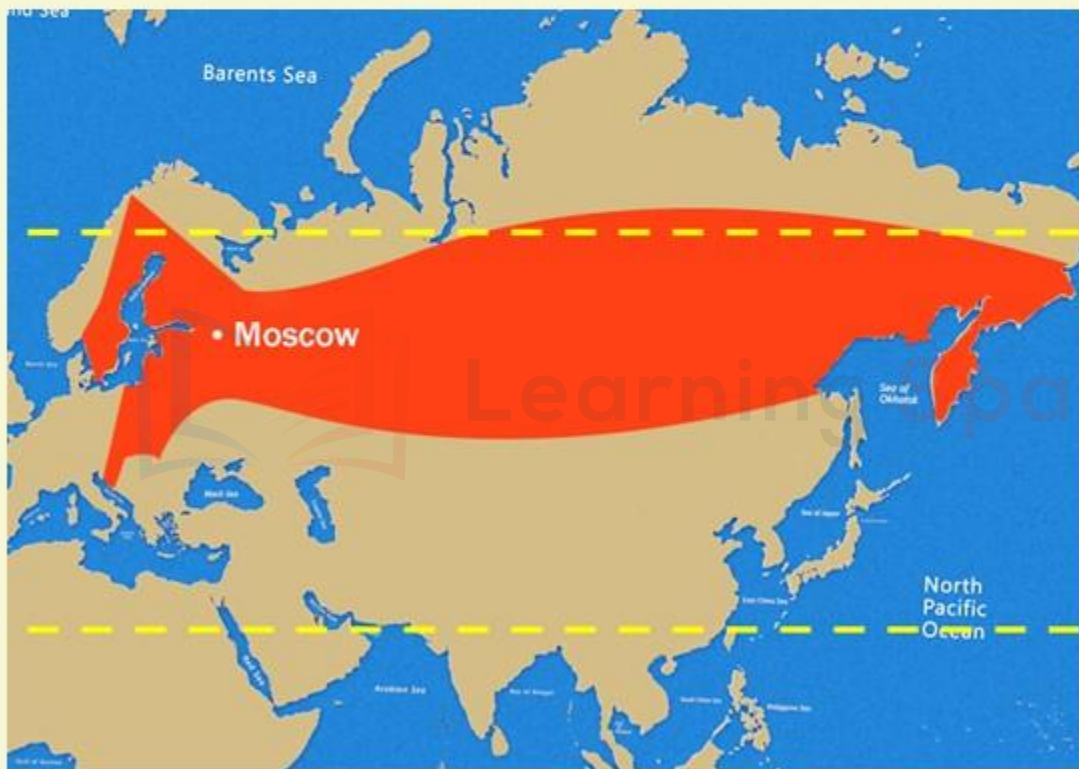
8 Occasionally cold, northerly polar winds such as the blizzards of Canada and Buran of Eurasia blow violently at 50 m.p.h or more and at a temperature of 10° C below freezing-point.

9 Conditions are so unbearable that Siberia is very sparsely populated but it is gradually being developed.

7 THE SIBERIAN CLIMATE

CLIMATE

b PRECIPITATION



- 1 The interiors of the Eurasian continent are **so remote from maritime influence** that annual precipitation cannot be high.
- 2 Generally, speaking, a total of **40-60 cm** is typical of the **annual precipitation** of this sub-arctic type of climate.
- 3 It is quite well **distributed** throughout the year, with a **summer maximum** from convectional rain.

7 THE SIBERIAN CLIMATE

NATURAL VEGETATION



- 1 No other trees are so well adapted as the **conifers** to withstand such an inhospitable environment as the Siberian type of climate.
- 2 The coniferous forest belts of Eurasia and North America are the **richest sources of softwood** for use in building construction, furniture, matches, paper and pulp, rayon and other branches of chemical industry.
- 3 The world's greatest softwood producers are USSR, USA, Canada, Finland, Norway and Sweden.

7 THE SIBERIAN CLIMATE

NATURAL VEGETATION

4 In the production of wood pulp (by both chemical and mechanical methods), the USA is the leader.

5 But, in the field on newsprint, Canada has outstripped all other producers, accounting for almost half of the world's total annual production.

6 There are four major species in the coniferous forests.

- 1) Pine
- 2) Fir e.g. Douglas fir & balsam fir.
- 3) Spruce
- 4) Larch

7 Their presence in pure stands and the existence of only a few species are a great advantage in commercial forest exploitation.

7**THE SIBERIAN CLIMATE****NATURAL VEGETATION****ADAPTATION OF CONIFERS****1****CONIFEROUS FORESTS ARE OF MODERATE DENSITY**

Unlike the equatorial rain forests which are luxuriant and contain trees of various heights, the coniferous forests are more uniform.

3**ALMOST ALL CONIFERS ARE EVERGREEN**

There is no annual replacement of new leaves as in deciduous trees. The same leaf remains on the tree for as long as five years.

2**CONIFERS ARE CONICAL IN SHAPE**

This is another adaption to survive the sub-Arctic climate. The sloping branches prevent snow accumulation which may snap the branches.

4**LEAVES ARE SMALL, THICK, LEATHERY AND NEEDLE-SHAPED**

This is to check excessive transpiration.

7 THE SIBERIAN CLIMATE

NATURAL VEGETATION

ADAPTATION OF CONIFERS

5 THERE IS LITTLE UNDERGROWTH

- The soils of the coniferous forests are poor. They are excessively leached and very acidic.
- The evergreen leaves provide little leaf-fall for humus formation, and the rate of decomposition of the leathery needles in a region of such low temperature is slow.
- Absence of direct sunlight and the short duration of summer are other contributory factors to a sparse undergrowth.
- All these factors are deterrents to the growth of much undergrowth.

7

THE SIBERIAN CLIMATE

ECONOMIC DEVELOPMENT

1 The coniferous forest regions of the northern hemisphere are comparatively little developed.

2 In Canada, eastern Europe, Asiatic Russia, large tracts of coniferous forests are still untouched.

3 Only in the more accessible areas are the forests cleared for lumbering.

4 There is little agriculture, as few crops can survive in the sub-Arctic climate of these northerly lands.

5 Many of the Samoyeds and Yakuts of Siberia and some Canadians are engaged in hunting, trapping and fishing.

6 We shall deal with two of the major activities in greater detail.

7

THE SIBERIAN CLIMATE ECONOMIC DEVELOPMENT

TYPE- 1

TRAPPING



- 1 To ensure a more regular supply of furs many fur farms have been established in Canada.
- 2 They fetch high prices in sophisticated cities like New York, London, Paris, Rome, Zurich, where pelts are processed as attractive fur coats and women's handbags
- 3 As in Canada, fur-farming has now replaced hunting of wild animals in many parts of Siberia as the main source of furs.

7

THE SIBERIAN CLIMATE ECONOMIC DEVELOPMENT

TYPE- 2

LUMBERING



- 1 That is probably the most important occupation of the Siberian type of climate.
- 2 The vast reserves of coniferous forest provide the basis for the lumbering industry.
- 3 **PAPER AND PULP INDUSTRY:**
 - Timber is pulped by both chemical and mechanical means to make wood pulp the raw material for paper-making and newsprint.
 - The development of the printing industry has made paper and pulp indispensable.
 - Canada and USA are leading producers of the world's supplies of newsprint and wood pulp respectively.

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao
K. Srikanth
D. Sunil Kumar
L. V. Krishna
Y. Deepthi

Amrita Naidu
M. Binukrishna
G. Ravi Babu
D. Joji
K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 -2 4 4 4 4 7 2
0 9 8 4 9 9 4 2 2 9 9